

Call for Expressions of Interest	
<i>Date of Issue:</i>	September 2026
<i>Expected Completion Date:</i>	May 2027
<i>Project Title:</i>	TouchLab: Exploring How Humans Feel about Robots Touch
<i>Company:</i>	Please provide company name
Contact Details	
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<i>E-mail:</i>	Additional contact e-mail
Additional Info	
<i>Students must sign NDA:</i>	☐
<i>Project uses personal data:</i>	☐
<i>Custom hardware required:</i>	☐ If Applicable: Please provide details
<i>Relevant research areas (if applicable)¹:</i>	Click or tap here to enter text.
Project Background Information	
What does it feel like when a robot touches you? And how can we understand why one interaction feels interesting, comforting, or surprising, while another feels uncomfortable or unremarkable? Robots that physically interact with people are increasingly being developed for physically assistive care (e.g. moving, feeding, comforting humans) and socially assistive robot (e.g. companion robots), but creating touch that feels appropriate, comfortable, or meaningful remains a challenging problem. A robot's movement, force, timing, and contact with the body can all influence how an interaction is experienced. Understanding these relationships requires more than recording what the robot does: researchers also need to understand what happens in the person's body and how the person experiences the interaction. This project will develop TouchLab, an interactive software platform for exploring these relationships. Students will work hands-on with a wearable soft robotic sleeve and a Franka robotic arm provided by the lab, experimenting with different robotic interactions and seeing how changes in the robot's behaviour relate to human responses. The platform will connect	

¹ An overview of different research areas in the school of computer science can be found [here](#).

robotic actuation, biophysical sensing, and user feedback through a shared timeline, allowing users to mark meaningful moments and explore what the robot was doing at those points.

The project builds on the *Cabinet of Affective Touches*, which supports retrospective descriptions of overall experience. TouchLab complements it by capturing moment-by-moment feedback, helping researchers investigate how sensory qualities emerge and change over time.

H. Sambi, C. Y. Zheng, P. Tennent, Y. Luft, S. Benford, and M. Balaam, 'A Cabinet of Affective Touches: Building Design Repository Linking Somatic Experience with Soft Robotic Systems', in Proceedings of the 2026 Conference on Creativity and Cognition, in C&C '26. New York, NY, USA: Association for Computing Machinery, Jul. 2026, pp. 497–511. doi: 10.1145/3803784.3809294.

Requirements

The group must provide:

The student group must design and implement a working, extensible software platform for capturing and exploring multimodal human–robot interaction data.

The platform must provide:

- **Modular data acquisition:** Interfaces for receiving data from multiple sensors, actuators, and robotic systems.
- **Time synchronisation:** A common timestamped data model that preserves temporal relationships between streams with different sampling rates.
- **User feedback:** An interface for recording interaction sessions and capturing time-referenced feedback through voice and/or explicit moment marking.
- **Live visualisation:** A shared timeline showing robotic parameters, sensor data, and user annotations.
- **Replay and analysis:** Facilities to replay recorded sessions, inspect selected moments, and export synchronised data for research analysis.
- **Interactive exploration and two-system evaluation:** The group must test and demonstrate the platform using a wearable soft robotic prototype and a Franka robotic arm provided by the lab. The demonstration should be engaging and hands-on, allowing users to experience the robotic interactions, explore different behaviours, and investigate how their feedback relates to the recorded data. It should also demonstrate that the platform can accommodate different robotic systems through its modular interfaces

Skills	
<i>Highly desirable:</i> • <i>Programming experience</i> • <i>Interest in user-facing software</i> • <i>Software Design and Development</i> • <i>Experience with data structures and data handling</i>	<i>Desirable:</i> • <i>Physical computing</i> • <i>Experience with programming hardware</i> • <i>Willingness to learn unfamiliar technologies</i> • <i>User Studies or social experiments</i>
Submission / Award information	
<i>Date of Submission:</i>	September 2026
<i>Date of Pitch:</i>	October 2026
<i>Notification of award:</i>	October 2026

For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.